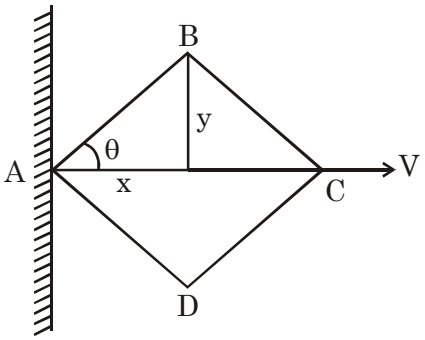
SOLUTION
SECTION-I

Single Correct Answer Type **10 Q. [3 M (–1)]**

1. Ans. (C)

Sol.



$AC = 2x$

$$\frac{d(AC)}{dt} = 2\frac{dx}{dt}$$

$$v = 2\frac{dx}{dt}$$

$x^2 + y^2 = \ell^2$

$$2x\frac{dx}{dt} + 2y\frac{dy}{dt} = 0$$

$$\frac{dy}{dt} = -\left(\frac{x}{y}\right) \times \frac{dx}{dt}$$

When rhombus is a square $\Rightarrow x = y$

$$V_B = -\frac{V}{2}$$

Speed of B = $\frac{V}{2}$

2. Ans. (D)

Sol. For the semicricular lamina of mass m, the moment of inertia about an axis through C is $I_C = \frac{1}{2}mr^2$.

Let I_{CM} = moment of inertia about an axis through its centre of mass.

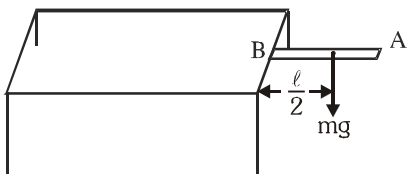
$$I_C = I_{CM} + md^2 \Rightarrow I_{CM} = \frac{1}{2}mr^2 - m\left(\frac{4r}{3\pi}\right)^2 = \left(\frac{1}{2} - \frac{16}{9\pi^2}\right)mr^2$$

3. Ans. (A)

Sol. Torque acting about the point B = $(mg)\frac{\ell}{2}$

If I is the moment of inertia of the stick about B and α is the angular acceleration then

$$I\alpha = (mg)\frac{\ell}{2} \Rightarrow \alpha = \frac{mg\ell}{2I}$$



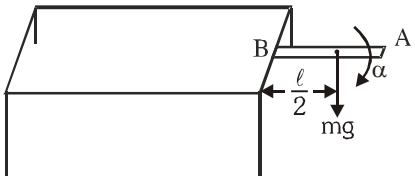
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Acceleration of centre of mass $a = \alpha\left(\frac{\ell}{2}\right) = \frac{mg\ell^2}{4I}$

Moment of inertia of this stick about its edge $I = \frac{m\ell^2}{3}$

$\Rightarrow a = \frac{mg\ell^2}{4\left(\frac{m\ell^2}{3}\right)} = \frac{3}{4}g$

OR



For angular motion of the stick

$\tau = mg\left(\frac{\ell}{2}\right) = I\alpha$

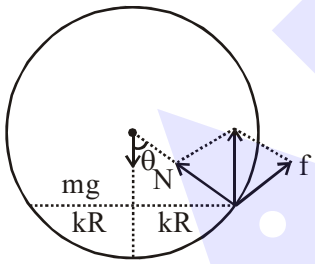
moment of intertia of stick about B is $I = \frac{m\ell^2}{3}$

$\Rightarrow mg\left(\frac{\ell}{2}\right) = \left(\frac{m\ell^2}{3}\right)\alpha \Rightarrow \alpha = \frac{3g}{2\ell}$

Acceleration of centre of mass $= \alpha\left(\frac{\ell}{2}\right) = \left(\frac{3g}{2\ell}\right)\left(\frac{\ell}{2}\right) = \frac{3}{4}g$

4. Ans.(B)

Sol.



$mg\cos\theta = N$
$f = mg\sin\theta < \mu N$
$\mu > \tan\theta$
$\sin\theta = k$

$\mu > \frac{k}{\sqrt{1-k^2}}$

5. Ans.(D)

Sol. $\omega_{sec} = \frac{2\pi}{60}$ rad/sec.

$\omega_{min} = \frac{2\pi}{3600}$ rad/sec.

$\omega_{hour} = \frac{2\pi}{43200}$ rad/sec.

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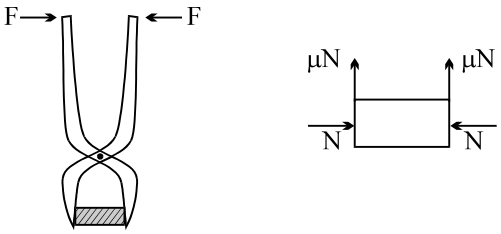
$\alpha = 0$ for all $\Rightarrow R\alpha = 0$ for all

$\theta = 30°$ after every 1 hr. and 5 min

6. Ans. (D)

Sol. $2\mu N = 2000$

$\therefore N = 10000$ N



also $F \times 30 = N \times 15$

$\Rightarrow F = 5000$ N

7. Ans. (C)

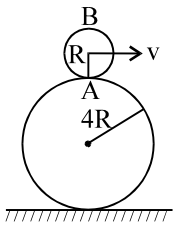
Sol. For toppling $mg\sin\theta\frac{3R}{8} \geq mg\cos\theta R$

$$\tan\theta \geq \frac{8}{3}$$

8. Ans.(A)

9. Ans. (A)

Sol.



$$\vec{a}_A = \vec{a}_{A/cm} + \vec{a}_{cm}$$

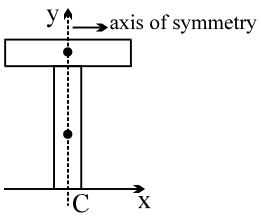
$$\vec{a}_{cm} = \frac{v^2}{5R}\downarrow$$

$$\vec{a}_{A/cm} = R\omega^2 = \frac{v^2}{R}\uparrow$$

$$\vec{a}_A = \left(\frac{v^2}{R} - \frac{v^2}{5R}\right)\uparrow = \frac{4v^2}{5R}\uparrow$$

10. Ans. (C)

Sol. Only translation is possible if force is applied at center of mass



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$$cm = \frac{2m.l + m.2l}{3m} = \frac{4}{3}l$$

Multiple Correct Answer Type

5 Q. [4 M (–1)]

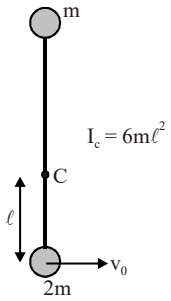
11. Ans.(BC)

Sol. For (B) : $2mv_0\ell = 6m\ell^2\,\omega$

$$\therefore\ \omega = \frac{v_0}{3\ell}$$

$$T = 2m\omega^2\ell = \frac{2mv_0^2}{9\ell}$$

For (C) : $I_A = 2m(3\ell)^2$; $I_B = m(3\ell)^2$



12. Ans.(B, C)

Sol. Balancing torque about support,

$85g \times 1.5 + 15\,g \times 0.5 = T \times 0.5$

$\Rightarrow T = 2700$ N

Balancing force in vertical direction,

$N = 15g + 85g + 2700 = 3700$

13. Ans.(A,C)

Sol. $f_1 = N_2$

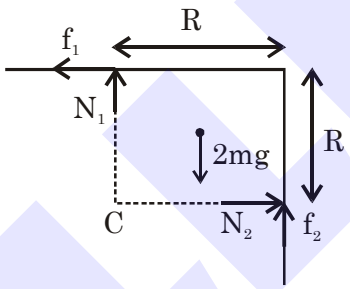
$N_1 + f_2 = 2mg$

$f_1 = f_2 = f$

$N_1 + f = 2\,mg$

$f = N_2$

$(\vec{\tau}_{net})_C = 0$



$$2fR - 2mg\frac{R}{2} = 0$$

$$f = \frac{mg}{2}$$

$$N_1 = 2mg - f = \frac{3mg}{2}$$

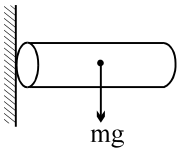
14. Ans.(A,C)

Sol. (A) $T_{mg} = I_\alpha$

$$Mg \times \frac{1}{2} = \frac{ML^2}{3}\alpha \text{ i.e. } \alpha = 3g/2L$$

(B) $a = r\alpha$

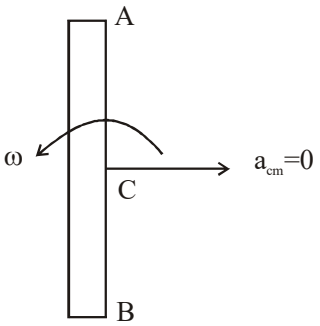
$$a = L\alpha = \frac{3g}{2}$$



15. Ans.(A,B,C,D)

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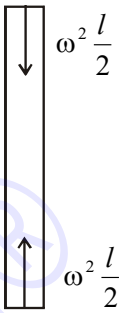
Sol.



$$\omega = \frac{11+3}{7/2} = 4$$

So accelerations of A & B is

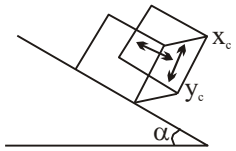
$$|a_A| = |a_B| = \omega^2 \frac{L}{2} = 4^2 \times \frac{7}{2} = 56$$



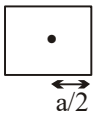
Linked Comprehension Type (1 Para × 3Q.) (1 Para × 2Q.) [3 M (-1)]
(Single Correct Answer Type)

16. Ans.(A)
17. Ans.(A)
18. Ans.(B)

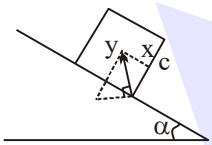
Sol. (i)



at toppling cm is just a box base line.



$$x_c = \frac{\frac{a}{2}\times m + \frac{a}{2}\times m + 0\times m}{3m} = \frac{a}{3}$$



$$y_{cm} = \frac{a}{2}$$

$$\tan(90^\circ - \alpha) = \frac{y_c}{x_c} = \frac{3}{2} \Rightarrow \tan\alpha = \frac{2}{3}$$

(ii) for it to not slide $3mg\sin\alpha \le 3\mu mg\cos\alpha$

$$\tan\alpha \le \mu$$

$$\frac{2}{3} \le \mu$$

$$\Rightarrow \mu \ge 0.66$$

(iii) $a_n = g\sin\alpha$

$$\Rightarrow a = \frac{1}{2} g\sin\alpha t^2 \qquad \Rightarrow t = \sqrt{\frac{2a}{g\sin\alpha}}$$

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Paragraph for Question no. 19 and 20

19. Ans.(D)

20. Ans.(D)

SECTION-II

Numerical Answer Type Question **2Q.[3M(0)]**

(upto second decimal place)

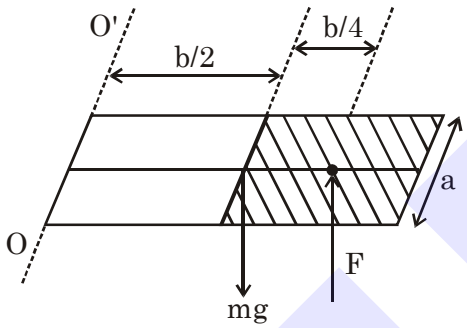
21. Ans. 500.00

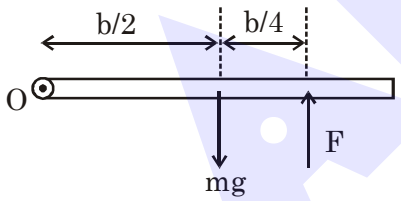
Sol. (a) The net torque that the rope exerts on the capstan, and hence the net torque that the capstan exerts on the rope, is the difference between the forces of the ends of the rope times the radius of the capstan. The capstan is doing work on the rope at a rate $P = \tau\omega = F_{net}\, r\, \omega = 500 \times (5.0 \times 10^{-2}\ m) \times 10 = 250$ W. A larger number of turns might increase the force, but for given forces, the torque is independent of the number of turns.

(b) $\frac{\Delta T}{t} = \frac{Q/t}{mc} = \frac{P}{mc} = \frac{250}{5\times1000} = \frac{1}{20} = 0.05 = 500 \times 10^{-4}\,°C/s$

22. Ans. 10 m/s

Sol.





Taking torque about axis OO' & putting it equal to zero

$(\vec{\tau}_{net})_0 = 0$

$mg\frac{b}{2} - F\frac{3b}{4} = 0$

$F = \frac{2mg}{3}$ .... (i)

$F = n(2mv)\left(\frac{ab}{2}\right)$ .... (ii)

From (i) & (ii), putting the values we get

$v = 10$ m/s

SECTION-III

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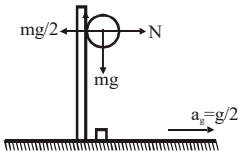
Numerical Grid Type (Ranging from 0 to 9) **2 Q. [4 M (0)]**

23. Ans.8

Sol. $\tau = I\alpha$; $\tau = mg\left(\frac{4R}{3\pi}\right)$; $I = \frac{MR^2}{2}$, $\alpha = \frac{8g}{3\pi R} \Rightarrow x = 8$

24. Ans.8

Sol.



For the given sphere :

$mg - f_s = m(a_{cm})_{rel}$ ...(1)

$N = \frac{mg}{2}$; $f_s \times R = \frac{2}{5}mR^2\alpha$; $a = \alpha R$

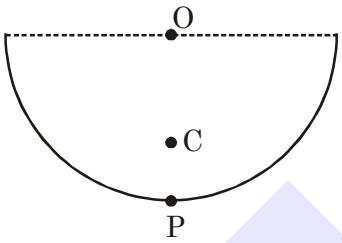
SECTION-IV

Matrix Match Type (4 × 5) **3 Q. [8 M (for each entry +2(0)]**

25. Ans. (A) -Q (B)-P (C)-S (D)-R

Sol. (A) $I = \frac{ML^2}{3}\sin^2 30° = \frac{ML^2}{12}$

(B)



$I_O = MR^2$

$I_O = I_C + M\left(\frac{2R}{\pi}\right)^2$

$I_P = I_C + M\left(R - \frac{2R}{\pi}\right)^2$

$I_P - I_0 = -\frac{3}{11}MR^2$

$I_P = \frac{8}{11}MR^2$

(C) $I_{base} = \frac{Mh^2}{6} = \frac{M}{6}\left(\frac{R\sqrt{3}}{2}\right)^2 = \frac{MR^2}{8}$

(D) $I = I_{(+)} + I_{(-)}$

$= \frac{M(2R)^2}{2} - \frac{M}{4}\left(\frac{R^2}{2} + R^2\right)$

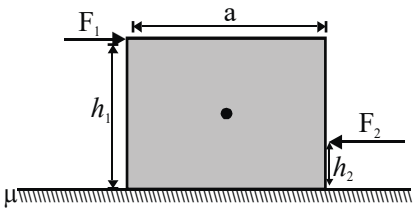
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$$= 2MR^2 - \frac{M}{8}3R^2$$

$$= \frac{13}{8}MR^2$$

26. Ans.(A) PT (B) PR (C) PQS (D) R

Sol.



(A) $F_1 = F_2$, So friction = 0

Since both torques of F_1 and F_2 act clockwise, so N has to shift rightwards to balance.

About COM : $F_1\frac{a}{2} + F_2\left(\frac{a}{2} - \frac{a}{8}\right) = N \cdot x$

$$\frac{mg}{2}\frac{a}{2} + \frac{mg}{2}\cdot\left(\frac{3a}{8}\right) = mg \cdot x$$

$$\frac{mga}{4} + \frac{3mga}{16} = mgx \Rightarrow x = \frac{7a}{16}$$

Since $x < \frac{a}{2}$

so this shift of normal is possible, so no toppling

(B) $F_2 > F_1$, so friction acts rightwards and its torque is out of the plane (about COM)

- $fr = F_2 - F_1 = \frac{mg}{2}$
- Normal shifts right for torque balance
- Balancing the torque about COM :

$$\frac{mg}{2}\cdot\frac{a}{2} + mg \times \left(\frac{a}{2} - \frac{a}{8}\right) = fr \times \frac{a}{2} + N \times x$$

$$\frac{mga}{4} + mg \cdot \frac{3a}{8} = \frac{mg}{2}\cdot\frac{a}{2} + mgx$$

$$\frac{5mga}{8} = \frac{mga}{4} + mgx$$

$$\frac{3mga}{8} = mgx$$

$$x = \frac{3a}{8}$$

Since $x < \frac{a}{2}$

so this shift of normal is possible, so no toppling

(C) $F_1 > F_2 \therefore$ friction acts leftwards and its torque is into the plane of paper

Torque of F_1 and friction is clockwise, so normal shifts to right

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Balancing the torque :

$$mg\times\frac{a}{2}+mg\times\frac{a}{2}=N\times x$$

$mga = mgx$

$x = a$

Since $x > \frac{a}{2}$

so this shift of normal is not possible, so toppling will take place

(D) $F_2 > F_1$, so friction acts rightwards and its torque is out of the plane

$$fr = F_2 - F_1 = \left(mg-\frac{mg}{2}\right)=\frac{mg}{2}$$

Since torque of F_1 & $F_2 = 0$

and only torque of friction acts anticlockwise

So normal has to shift left to avoid toppling.

About COM :

$$\frac{mg}{2}\frac{a}{2}=N\times x=mgx$$

$$x=\frac{a}{4}$$

Since $x < \frac{a}{2}$

so this shift of normal is possible, so no toppling

27. Ans.(A) → (R) ; (B) → (S) ; (C) → (P) ; (D) → (Q)

Sol. **For (A) :** $I = MR^2 + 3\left(\frac{MR^2}{3}\right) = 2MR^2 \Rightarrow K = \sqrt{\frac{I}{M}} = \sqrt{\frac{2MR^2}{4M}} = \frac{R}{\sqrt{2}}$

For (B) : $I = MR^2 + 4\left(\frac{MR^2}{3}\right) = \frac{7MR^2}{3} \Rightarrow K = \sqrt{\frac{7MR^2}{3(5M)}} = R\sqrt{\frac{7}{15}}$

For (C) : $I = MR^2 + \frac{1}{2}MR^2 = \frac{3MR^2}{2} \Rightarrow K = \sqrt{\frac{3MR^2}{2(2M)}} = \frac{\sqrt{3}}{2}R$

For (D) : $I = MR^2 + \frac{1}{2}MR^2 + \frac{1}{3}MR^2 = \frac{11}{6}MR^2 \Rightarrow K = \sqrt{\frac{11MR^2}{6(3M)}} = R\sqrt{\frac{11}{18}}$

Subjective Type

3 Q. [4 M (0)]

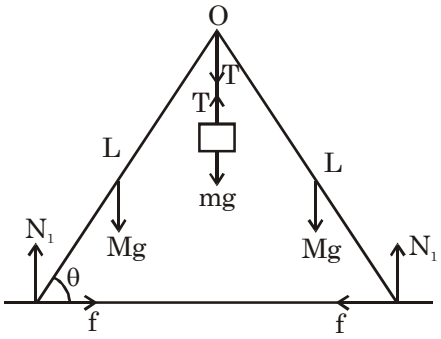
28. Ans. $f = (M+m)\,g\,\frac{\cot\theta}{2}$

Sol. For any one ladder

$(\vec{\tau}_{net})_0 = 0$

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$$N_1 L\cos\theta - f L\sin\theta - Mg\frac{L}{2}\cos\theta = 0$$



$$\vec{F}_{net} = 0$$

$$2N_1 - 2Mg - mg = 0$$

$$N_1 = \left(M + \frac{M}{2}\right)g$$

$$f = \cot\theta\left[N_1 - \frac{Mg}{2}\right]$$

$$f = \cot\theta\left[M + \frac{m}{2} - \frac{M}{2}\right]g$$

$$f = \frac{(M+m)g(\cot\theta)}{2}$$

29. **Ans.** $\frac{a\ell^4}{36}$

Sol. $I_0 = \int dmx^2 = \int_0^{\ell} \lambda dx x^2 = \frac{a\ell^4}{4}$

Now CM = **d** = $\frac{\int_r^{\ell} \lambda dx x}{\int_0^{\ell} \lambda dx} = \frac{2\ell}{3}$

$$\therefore I_{CM} = I_0 - md^2 = \frac{a\ell^4}{4} - md^2$$

$$m = \int \lambda dx = \frac{a\ell^2}{2}$$

$$\therefore \frac{a\ell^4}{4} - \left(\frac{a\ell^2}{2}\right)\frac{4}{9}\ell^2 = \frac{a\ell^4}{36}$$

30. **Ans.**(a) 5 m/s^2 $\rightarrow$, (b) $0.3 < h < 1.5$ m

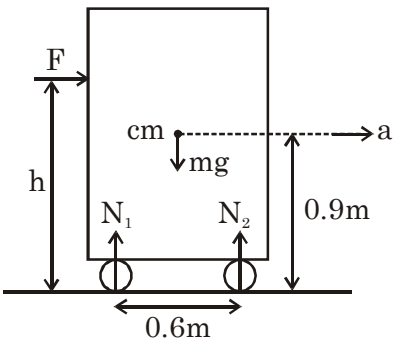
Sol. $F = ma$

$a = 5 m/s^2$

$N_1 + N_2 = mg = 200$ .... (i)

$(\vec{\tau}_{net})_{cm} = 0$

Case-1 : $F(h - 0.9) + N_1(0.3) - N_2(0.3) = 0$



$$\frac{100(h-0.9)}{0.3} = -(N_1 - N_2)$$

$$N_2 - N_1 = \frac{1000}{3}(h - 0.9) \quad \text{.... (ii)}$$

From (i) & (ii)

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$$2N_1 = 200 - \frac{1000}{3}(h - 0.9)$$

$$N_1 = 100 - \frac{500}{3}(h - 0.9)$$

For not tipping, $N_1 > 0$

$$\frac{300}{500} > h - 0.9$$

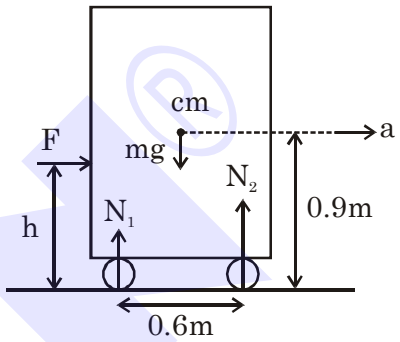
$$h < 0.9 + 0.6$$

$$h < 1.5 \text{ m}$$

$$(\vec{\tau}_{net})_{cm} = 0$$

Case-2 : $F(0.9 - h) + N_1(0.3) - N_2(0.3) = 0$ .... (iii)

From (i) & (iii)



$$N_1 = 100 - \frac{500}{3}(0.9 - h)$$

For not tipping, $N_1 > 0$

$$\frac{300}{500} > 0.9 - h$$

$$h > 0.9 - 0.6$$

$$h > 0.3 \text{ m}$$

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